Vol 2 Chapter 1 — Why Particle Physics from D_IV^5

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Chapter 1 — Why Particle Physics from D_{IV}^5

Volume 1 built the framework — Hilbert space, integers, operators, dynamics, gauge theory, renormalization. This volume applies the framework. By the end of Volume 2 the reader will have seen the substrate-derivation of the major Standard Model observables: gauge group structure, three fermion generations, color and quark confinement, the lepton sector, the proton-to-electron mass ratio (Chapter 6's recruiter result), CKM mixing, coupling constants including the electron's anomalous magnetic moment a_e, the Higgs sector, neutrino mixing, the five-absence prediction set, and the substrate-engineering experimental program.

Each chapter follows the same pattern: a measured Standard Model quantity, identified with a structural quantity on D_{IV}^5, derived to sub-percent precision with no fitted parameters, tier-classified honestly.

1.1 What this volume covers

Chapter	Sector	Headline result
2	SM Gauge Group	$SU(3) \times SU(2) \times U(1)$ forced from BST primaries; total dim 12 = $N_c \cdot \text{rank} \cdot 2$
3	Three Generations	Q^5 cohomology truncation; m_μ/m_e at 0.004% (Cal #100 corrected)
4	Color and Quarks	Frobenius confinement on D_{IV}^5; substrate trefoil topology
5	Lepton Sector	$m_\tau/m_e = 49 \cdot 71$ at 0.7%
6	m_p/m_e	$6\pi^5$ at 0.002% — Crown Jewel (rewritten)
7	CKM Mixing	Jarlskog J at 0.3% (T1444 vacuum-subtraction conditional)
8	Coupling Constants	a_e at ppt precision — second Crown Jewel
9	Higgs	m_H at 0.07–0.11% dual-route (partial derived, multi-month)
10	Neutrinos	Seesaw at BST primary 17
11	Five Absences	No GUT/proton decay/monopoles/sterile-ν/SUSY
12	Experimental Program	SP-29/SP-30 substrate-engineering falsifiers

Most of the chapters above have D-tier ratified derivations. A few (Higgs, parts of CKM) sit at I- or PARTIAL-DERIVED with honest scope; the chapters mark this explicitly per the Quaker discipline of Volume 0 Chapter 10.

1.2 Reader prerequisites

Volume 0 (substrate foundation) and Volume 1 (QFT machinery) are the load-bearing prerequisites. A reader who has worked through Volume 0 Chapter 9 (Strong-Uniqueness) and Volume 1 Chapters 2 (Hilbert space) and 8 (Gauge Theory) has the apparatus to follow Volume 2 directly.

Standard graduate particle physics — Peskin–Schroeder or Halzen–Martin level — is useful background but not required. Where this volume's substrate derivations parallel standard-textbook treatments, we will cross-reference.

1.3 What this volume claims

The framework's central claim, applied to particle physics: every Standard Model observable is structurally derivable from D_{IV}^5 and the five BST primaries, with no fitted parameters. The match precisions span 0.002% to 0.7% across roughly a hundred independent quantities. The substrate produces the Standard Model; the Standard Model is not an empirical input.

A reader who finishes this volume should be able to articulate, for each Standard Model observable they care about, the substrate-mechanism reading and the BST-primary integer combination that produces it.

1.4 What comes next

Chapter 2 derives the gauge group $SU(3)\times SU(2)\times U(1)$ in operational detail, picking up from Volume 1 Chapter 8's substrate-mechanism framework.

Where to look this up: Per-chapter T-numbers and verification toys as cited. The `verify_bst.py` reproduction script covers the volume's load-bearing observables.